

PROJECT TITLE

**OPEN SOURCE INTELLIGENCE
(OSINT) INVESTIGATION ON OPENAI**

Submitted By

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Course: Cyber Security Internship

EXECUTIVE SUMMARY

Open Source Intelligence (OSINT) is one of the most important techniques used during cybersecurity investigations, penetration testing, and digital forensics. It involves collecting publicly available information from various online sources without directly interacting with the target organization. Such information assists security professionals in understanding an organization's digital footprint, identifying publicly exposed assets, and assessing potential security risks.

This report presents an OSINT investigation conducted on **OpenAI**, a leading artificial intelligence research and deployment company. The investigation was performed using ethical and publicly available intelligence-gathering techniques. Various OSINT tools, including Google Search, WHOIS, Google Dorking, Shodan, and Maltego, were used to collect information related to the organization's domains, publicly available contact information, social media accounts, technology stack, and internet-facing assets.

The collected information was analyzed to identify possible information exposure risks and understand how publicly available data can support both cybersecurity professionals and potential attackers during reconnaissance. Finally, a relationship diagram was developed using Maltego, and a risk assessment was performed based on the findings.

INTRODUCTION

Open Source Intelligence (OSINT) refers to the systematic process of collecting, analyzing, and interpreting information that is publicly available from various sources such as websites, search engines, public databases, social media platforms, domain registration records, and internet-connected services. Unlike intrusive security assessments, OSINT relies solely on legally accessible information without exploiting vulnerabilities or gaining unauthorized access to systems.

In the field of cybersecurity, OSINT plays a crucial role during the reconnaissance phase of security assessments. Security analysts, penetration testers, digital forensic investigators, and threat intelligence teams use OSINT techniques to gather information about organizations, identify publicly exposed assets, monitor digital footprints, and evaluate potential attack surfaces.

However, publicly available information can also be exploited by malicious actors to prepare phishing campaigns, perform social engineering attacks, fingerprint technologies, identify exposed services, and gather intelligence before launching cyberattacks. Therefore, organizations should continuously monitor the information they expose on the internet and implement appropriate security controls to reduce unnecessary exposure.

For this investigation, OpenAI has been selected as the target organization due to its significant online presence and the availability of sufficient publicly accessible information required to demonstrate various OSINT techniques. The investigation focuses exclusively on ethical intelligence gathering using publicly available sources without attempting to access any restricted information.

OBJECTIVES

The primary objectives of this OSINT investigation are:

- To collect publicly available information about OpenAI using ethical intelligence gathering techniques.
- To identify the official domains associated with the organization.
- To identify publicly available email addresses and contact information.
- To identify the official social media accounts maintained by the organization.
- To analyze the publicly visible technology stack used by the organization's website.
- To investigate internet-facing assets using publicly available search engines.
- To assess potential information exposure risks.
- To create a relationship diagram illustrating the connections between identified entities.
- To prepare a comprehensive intelligence report and risk assessment based on the investigation findings.

SCOPE OF INVESTIGATION

The scope of this investigation is limited to collecting information that is publicly accessible through legitimate OSINT techniques. No attempts were made to bypass authentication mechanisms, exploit vulnerabilities, or gain unauthorized access to any system.

The investigation includes:

- Collection of basic company information.
- Domain registration analysis.
- Public email investigation.
- Social media profiling.
- Technology stack identification.
- Internet-facing infrastructure analysis.
- Information exposure assessment.
- Relationship mapping using Maltego.

The investigation excludes:

- Vulnerability exploitation.
- Password attacks.
- Unauthorized scanning.
- Network intrusion.
- Access to confidential or restricted information.

INVESTIGATION METHODOLOGY

The investigation followed a structured OSINT methodology consisting of the following phases:

Phase 1: Target Identification

The target organization, OpenAI, was selected for investigation. Basic company information was collected to understand its business operations and online presence.

Phase 2: Public Information Collection

Publicly available information was collected using search engines and the official website. This included organizational details, headquarters, products, services, and publicly available contact information.

Phase 3: Domain Investigation

WHOIS lookup services were used to identify domain registration information, registrar details, domain status, registration dates, expiration dates, and DNS information.

Phase 4: Public Information Discovery

Google Dorking techniques were used to identify publicly indexed documents, contact pages, and publicly available resources associated with the organization.

Phase 5: Infrastructure Analysis

Technology fingerprinting tools and Shodan were used to identify publicly visible technologies and internet-facing services associated with the organization's web presence.

Phase 6: Intelligence Analysis

The collected information was analyzed to determine potential security implications, information exposure risks, and the overall digital footprint of the organization.

Phase 7: Relationship Mapping

Maltego was used to visualize the relationships among domains, public contact information, technology stack, social media accounts, and other identified entities.

TARGET PROFILE

The target selected for this investigation is **OpenAI**, an artificial intelligence research and deployment company headquartered in San Francisco, California, USA. The organization develops advanced AI models and applications for research, education, software development, and enterprise solutions. OpenAI is recognized globally for developing products such as ChatGPT, GPT models, DALL·E, Codex, and Sora.

The company maintains an extensive online presence through its official website, developer platform, documentation, social media accounts, and publicly accessible resources, making it a suitable target for demonstrating ethical OSINT investigation techniques.

COMPANY OVERVIEW

The initial stage of the investigation involved collecting publicly available information about OpenAI from trusted sources, including Google Search and the official website.

Procedure Performed

1. Opened a web browser and navigated to Google Search.
2. Searched for the keyword "**OpenAI**".
3. Opened the official OpenAI website.
4. Reviewed publicly available company information.
5. Recorded the organizational details required for the investigation.

Screenshot 1 :

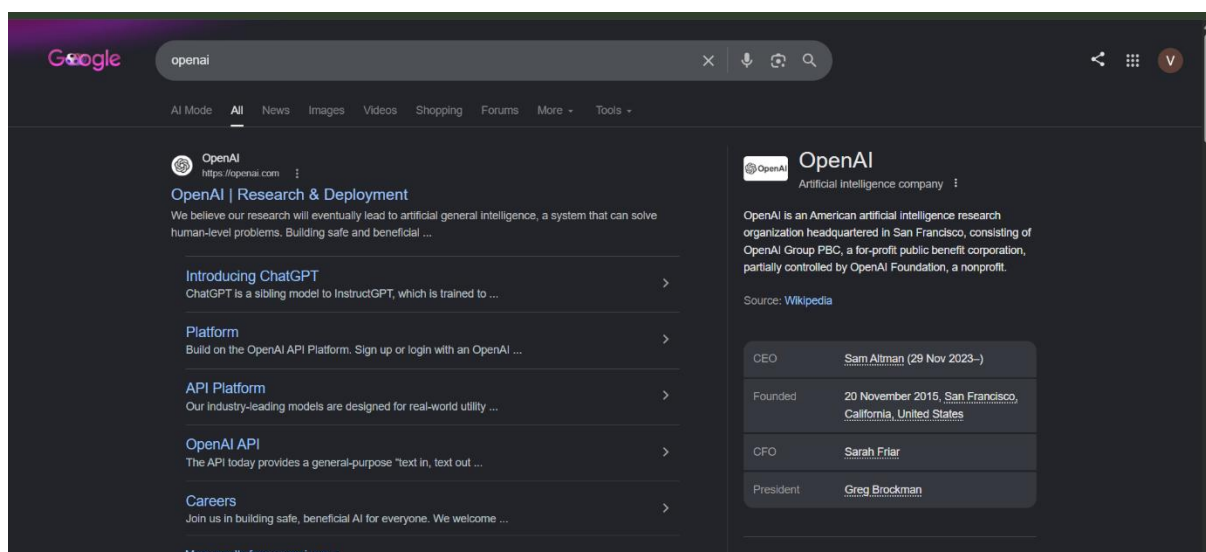


Figure 1. Google Search results displaying the official OpenAI website and company overview.

Screenshot 2:

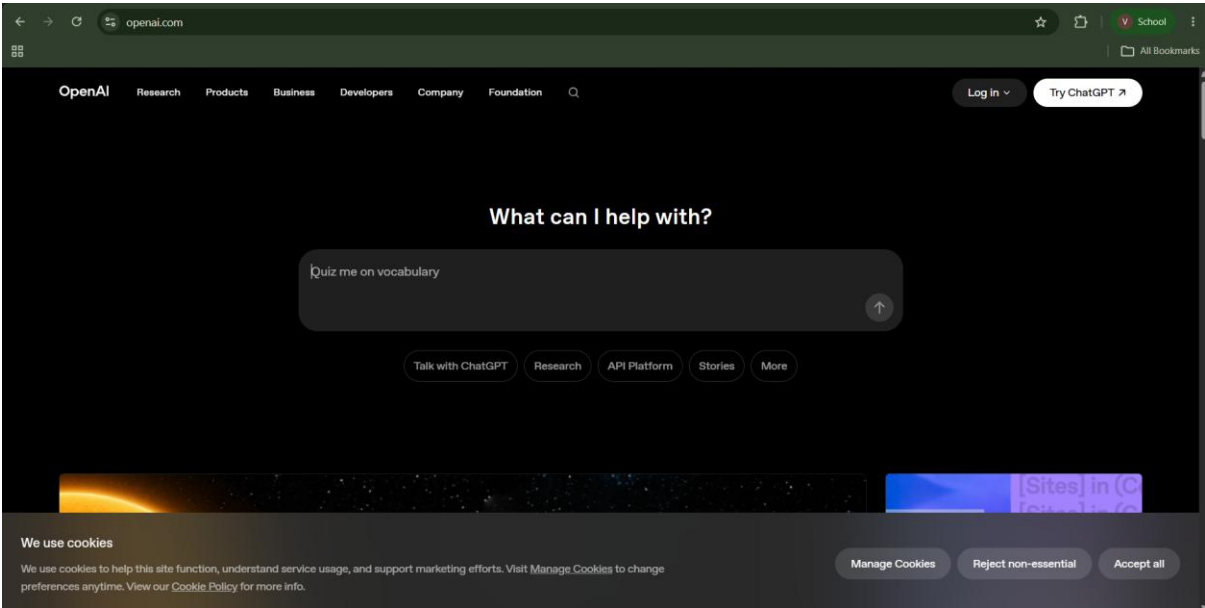


Figure 2. Official OpenAI website homepage.

COMPANY INFORMATION

Information	Details
Company Name	OpenAI
Company Type	Artificial Intelligence Research and Deployment Company
Official Website	https://openai.com
Headquarters	San Francisco, California, United States
Founded	20 November 2015
CEO	Sam Altman
President	Greg Brockman
CFO	Sarah Friar
Industry	Artificial Intelligence (AI)
Main Products	ChatGPT, GPT Models, OpenAI API, DALL·E, Codex, Sora

PUBLIC INFORMATION GATHERING

The first practical phase of the investigation involved gathering information that is publicly available through search engines and the official website.

The following information was collected:

Information Collected	Source
Official Website	https://openai.com
Headquarters	San Francisco, California, United States
CEO	Sam Altman
Founding Year	20 November 2015
Products and Services	ChatGPT, GPT Models, OpenAI API, DALL·E, Codex, Sora
Company Description	OpenAI is an artificial intelligence research and deployment company whose mission is to ensure that artificial general intelligence (AGI) benefits all of humanity. The organization develops advanced AI models and products, conducts AI research, and provides AI-powered services through applications such as ChatGPT and developer APIs. OpenAI focuses on building safe, reliable, and beneficial artificial intelligence for individuals, businesses, researchers, and developers

OBSERVATIONS

The Google search results indicate that OpenAI is a globally recognized artificial intelligence research and deployment company headquartered in **San Francisco, California, United States**. The organization was established in **2015** and is currently led by **Sam Altman** as Chief Executive Officer.

The search results provide direct links to the company's official website, developer platform, API documentation, ChatGPT portal, careers page, and research resources. These publicly available resources demonstrate that OpenAI maintains a strong online presence and offers extensive documentation regarding its products and services.

The information gathered during this phase establishes the identity of the target organization and serves as the starting point for further OSINT activities, including domain investigation, email discovery, technology stack analysis, and infrastructure reconnaissance.

INTELLIGENCE FINDING

The initial reconnaissance confirms that OpenAI has a significant public digital footprint. Essential organizational details, official web resources, leadership information, and product offerings are readily accessible through public search engines. This publicly available information provides valuable context for cybersecurity professionals during legitimate security assessments but could also be leveraged by attackers during the reconnaissance phase of a cyberattack. Therefore, organizations should ensure that publicly available information remains accurate, necessary, and does not unintentionally expose sensitive operational details.

ANALYSIS

The information collected through Google Search demonstrates the importance of public reconnaissance in an OSINT investigation. Even without using specialized tools, investigators can identify the target organization's official web resources, management information, products, and corporate structure. This information forms the baseline for deeper analysis using WHOIS, Google Dorking, Shodan, and Maltego in the subsequent phases of the investigation.

PRELIMINARY FINDINGS

Based on the initial investigation, the following observations were made:

- OpenAI maintains an official website that provides comprehensive information about its products, services, research, and developer resources.
- The company has a well-established online presence, making publicly available information readily accessible through search engines.
- Multiple AI products and services are publicly documented, indicating a mature digital ecosystem.
- The organization provides publicly accessible documentation and support resources that will be analyzed further in subsequent phases of the investigation.
- Additional intelligence regarding domains, public email addresses, technology stack, and internet-facing assets will be collected using specialized OSINT tools in the following sections of this report.

DOMAIN INVESTIGATION

After gathering the basic public information about OpenAI, the next phase of the investigation involved analyzing the organization's domain information. Domain investigation is an important OSINT technique that provides publicly available information about a website's registration details, ownership, registrar, DNS configuration, and domain status. This information helps security analysts understand the digital footprint of an organization and identify assets that may be useful during reconnaissance.

For this investigation, the domain **openai.com** was selected as the primary target because it is the official website of OpenAI.

OBJECTIVE

The objective of this investigation is to identify publicly available information related to the OpenAI domain using WHOIS lookup services.

The investigation aims to identify:

- Domain Name
- Registrar
- Registration Date
- Expiration Date
- Domain Status
- Name Servers
- DNS Information

TOOL USED

Tool Name: WHOIS Lookup

Purpose:

WHOIS is a publicly available internet database that stores domain registration information. It allows investigators to identify details related to domain ownership, registration history, registrar information, and DNS configuration.

Procedure

The following steps were performed during the investigation:

1. Open a web browser.
2. Visit a WHOIS lookup website (for example, ICANN Lookup or Whois.com).
3. Enter the domain name **openai.com** into the search bar.
4. Click **Search**.
5. Review the domain registration details.
6. Record the available information for further analysis.

Screenshot 3:

← → ↺ 📄 whois.domaintools.com/openai.com

🔍 School 📖 All Bookmarks

HOME RESEARCH

DomainTools

PROFILE CONNECT MONITOR SUPPORT

Whois Lookup 🔍

LOGIN Sign Up

Home > Whois Lookup > Openai.com

Notice: Possible deprecation of Whois services after January 26, 2025. [More Info](#)

How does this work?

Whois Record for Openai.com

Domain Profile

Registrar

MarkMonitor, Inc. MarkMonitor Inc.
IANA ID: 292
URL: http://www.markmonitor.com
Whois Server: whois.markmonitor.com
(p) +1.2086851750

Registrar Status

clientDeleteProhibited, clientTransferProhibited, clientUpdateProhibited, serverDeleteProhibited, serverTransferProhibited, serverUpdateProhibited

Dates

7,101 days old
Created on 2007-01-19
Expires on 2029-01-19
Updated on 2024-10-17

Name Servers

NS1-02.AZURE-DNS.COM (has 573,815 domains)
NS2-02.AZURE-DNS.NET (has 311 domains)
NS3-02.AZURE-DNS.ORG (has 162 domains)
NS4-02.AZURE-DNS.INFO (has 53 domains)

IP Address

104.18.33.45 - 80 other sites hosted on this server

IP Location

🇺🇸 California - San Jose - Cloudflare Inc.

ASN

🇺🇸 AS13335 CLOUDFLARENET - Cloudflare, Inc., US (registered Jul 14, 2010)

Domain Status

Registered And No Website

IP History

149 changes on 149 unique IP addresses over 21 years

Hosting History

5 changes on 5 unique name servers over 13 years

Whois Record (last updated on 2026-06-29)

Domain Name: openai.com
Registry Domain ID: 764064142_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2024-10-17T22:02:55+0000
Creation Date: 2007-01-19T19:28:24+0000
Registrar Registration Expiration Date: 2029-01-19T19:28:24+0000
Registrar: MarkMonitor, Inc.
Registrar IANA ID: 292
Registrar Abuse Contact: https://corp.markmonitor.com/domain/ui/abuse-report
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientUpdateProhibited (https://www.icann.org/epp#clientUpdateProhibited)
Domain Status: clientTransferProhibited (https://www.icann.org/epp#clientTransferProhibited)
Domain Status: clientDeleteProhibited (https://www.icann.org/epp#clientDeleteProhibited)
Domain Status: serverUpdateProhibited (https://www.icann.org/epp#serverUpdateProhibited)
Domain Status: serverTransferProhibited (https://www.icann.org/epp#serverTransferProhibited)
Domain Status: serverDeleteProhibited (https://www.icann.org/epp#serverDeleteProhibited)
Registrant Organization: OpenAI
Registrant Country: US
Registrant Email: Select Request Email Form at https://domains.markmonitor.com/whois/openai.com
Tech Email: Select Request Email Form at https://domains.markmonitor.com/whois/openai.com
Name Server: ns2-02.azure-dns.net
Name Server: ns4-02.azure-dns.info
Name Server: ns1-02.azure-dns.com
Name Server: ns3-02.azure-dns.org
DNSSEC: unsigned
URL of the ICANN WHOIS Data Problem Reporting System: http://wdprs.internic.net/

For more information on WHOIS status codes, please visit:

DomainTools Iris

The gold-standard internet intelligence platform

Learn More

Preview the Full Domain Report

Tools

Hosting History

Monitor Domain Properties

Reverse IP Address Lookup

Network Tools

Available TLDs

General TLDs Country TLDs

The following domains are available through our preferred partners. Select domains below for more information. (3rd party site)

Taken domain.

Available domain.

Deleted previously owned domain.

Unrescued previously owned domain.

OpenAi.com

View Whois

OpenAi.net

View Whois

OpenAi.org

View Whois

OpenAi.info

View Whois

OpenAi.biz

View Whois

OpenAi.us

View Whois

Figure 3: WHOIS Lookup for openai.com

WHOIS INFORMATION

Parameter	Value
Domain Name	openai.com
Registry Domain ID	764064142_DOMAIN_COM-VRSN
Registrar	MarkMonitor Inc.
IANA ID	292
WHOIS Server	whois.markmonitor.com
Registrar URL	http://www.markmonitor.com
Registration (Creation) Date	19 January 2007
Last Updated Date	17 October 2024
Expiration Date	19 January 2029
Registrant Organization	OpenAI
Registrant Country	United States (US)
Name Servers	NS1-02.AZURE-DNS.COM NS2-02.AZURE-DNS.NET NS3-02.AZURE-DNS.ORG NS4-02.AZURE-DNS.INFO
Domain Status	clientUpdateProhibited clientTransferProhibited clientDeleteProhibited serverUpdateProhibited serverTransferProhibited serverDeleteProhibited
DNSSEC	Unsigned
IP Address	104.18.33.45
IP Location	San Jose, California, United States
Hosting Provider / ASN	Cloudflare Inc. (AS13335 – CLOUDFLARENET)

EXPLANATION OF WHOIS FIELDS

1. Domain Name

The domain name identified during the investigation is openai.com, which is the official web domain used by OpenAI to provide its AI products, services, documentation, and developer resources. This domain serves as the primary online identity of the organization.

2. Registrar

The WHOIS record shows that the domain is registered with MarkMonitor Inc. (IANA ID: 292). MarkMonitor is a well-known domain registrar that provides domain management and brand protection services for large organizations. The use of a reputable registrar indicates that OpenAI follows professional domain management practices.

3. Registration Date

The domain was created on 19 January 2007. This indicates that the domain has existed for many years and has a long-established online presence. Although OpenAI was founded later, the domain itself was registered earlier and is now officially owned and managed by the organization.

4. Last Updated

The WHOIS record indicates that the domain information was last updated on 17 October 2024. Regular updates to domain records demonstrate that the organization actively manages its domain registration and maintains current administrative information.

5. Expiration Date

The domain registration is valid until 19 January 2029. A long registration period reduces the risk of accidental domain expiration and reflects effective domain lifecycle management.

6. Domain Status

The WHOIS record contains several protective status codes, including:

- clientUpdateProhibited
- clientTransferProhibited
- clientDeleteProhibited
- serverUpdateProhibited
- serverTransferProhibited
- serverDeleteProhibited

These status codes help protect the domain from unauthorized updates, deletion, or transfer, thereby improving the security of the organization's internet assets.

7. Name Servers

The domain uses the following Microsoft Azure DNS name servers:

- NS1-02.AZURE-DNS.COM
- NS2-02.AZURE-DNS.NET
- NS3-02.AZURE-DNS.ORG
- NS4-02.AZURE-DNS.INFO

These name servers are responsible for translating the domain name into IP addresses, enabling users to access OpenAI's online services. Using multiple name servers improves availability, redundancy, and fault tolerance.

8. DNSSEC

The WHOIS record indicates that DNSSEC is Unsigned. DNSSEC (Domain Name System Security Extensions) helps protect DNS records against spoofing and cache poisoning attacks by providing cryptographic validation. In this record, DNSSEC is not enabled according to the available WHOIS information.

DOMAIN ANALYSIS

The WHOIS investigation confirms that openai.com is actively registered and professionally managed by OpenAI. The domain is registered through MarkMonitor Inc., a trusted enterprise registrar commonly used by large organizations for secure domain management.

The domain has been registered since 19 January 2007 and remains valid until 19 January 2029, demonstrating long-term maintenance and administrative oversight. The use of Microsoft Azure DNS name servers indicates that the organization's DNS infrastructure is managed using a reliable cloud-based DNS service. Additionally, the WHOIS record shows that the website's public IP address is associated with Cloudflare, indicating that Cloudflare is used to enhance website performance, availability, and security.

Several domain protection mechanisms, including client and server protection flags, have been enabled to prevent unauthorized modifications, deletion, or transfer of the domain. These security measures reduce the risk of domain hijacking and unauthorized administrative changes.

INTELLIGENCE FINDINGS

The following observations were made during the domain investigation:

- The official domain used by the organization is openai.com.
- The domain is registered with MarkMonitor Inc., an enterprise-grade domain registrar.
- The domain was created on 19 January 2007 and is registered until 19 January 2029.
- The domain information was last updated on 17 October 2024, indicating active maintenance.
- Four Microsoft Azure DNS name servers are configured to provide reliable DNS resolution.
- Multiple domain protection status codes are enabled to prevent unauthorized transfers, updates, and deletion.
- The public IP address is associated with Cloudflare, suggesting the use of CDN and security services.
- The WHOIS record reports DNSSEC as Unsigned.

INFORMATION EXPOSURE ANALYSIS

The WHOIS investigation revealed only publicly available registration information. Although this information is intended to improve transparency and internet administration, it can also provide useful intelligence during the reconnaissance phase of a cyberattack.

Public Information	Possible Risk
Domain Name	Identifies the organization's official online presence.
Registrar Information	Reveals the domain management provider used by the organization.
Registration Timeline	Indicates the age and maintenance history of the domain.
Name Servers	Can assist attackers in DNS reconnaissance and infrastructure mapping.
IP Address and Hosting Information	Reveals that Cloudflare is used as the hosting/security provider, which may assist in infrastructure fingerprinting.
Domain Status	Shows that several security mechanisms are enabled to protect the domain from unauthorized administrative changes.

SECURITY RECOMMENDATIONS

Based on the WHOIS investigation, the following recommendations are suggested:

- Continue using enterprise-grade domain management services such as MarkMonitor.
- Periodically review WHOIS information to ensure that registration details remain accurate and up to date.
- Monitor DNS records regularly for unauthorized changes or suspicious activity.
- Maintain domain protection mechanisms such as transfer, update, and deletion restrictions.
- Consider implementing DNSSEC, if appropriate, to provide additional protection against DNS spoofing and cache poisoning attacks.
- Regularly audit publicly exposed domain information as part of the organization's cybersecurity governance process.

EMAIL INVESTIGATION (GOOGLE DORKING)

After completing the domain investigation, the next phase of the OSINT investigation focused on identifying publicly available email addresses and other indexed resources associated with OpenAI. This was accomplished using **Google Dorking**, an advanced search technique that utilizes Google's search operators to discover information indexed by search engines.

Google Dorking helps investigators locate publicly accessible contact information, documents, directories, login pages, and other resources that may not be immediately visible through standard browsing. During this investigation, only ethical and publicly accessible search queries were used.

OBJECTIVE

The objectives of this investigation are:

- To identify publicly available email addresses related to OpenAI.
- To discover publicly accessible contact pages.
- To identify publicly indexed documents.
- To analyze whether sensitive information is exposed through search engines.

TOOL USED

Tool Name: Google Search (Google Dorking)

Purpose:

Google Dorking uses advanced search operators to locate publicly indexed information that may assist investigators in understanding an organization's digital footprint.

GOOGLE DORKS USED

The following Google Dorks were used during the investigation.

Google Dork	Purpose
site:openai.com	Lists pages indexed from the official website
site:openai.com "@openai.com"	Searches for publicly available email addresses
site:openai.com contact	Identifies official contact pages
site:openai.com filetype:pdf	Searches for publicly accessible PDF documents

Procedure

The following steps were performed during the investigation:

1. Open Google Search.
2. Enter each Google Dork individually.
3. Review the indexed search results.
4. Record publicly available information.
5. Capture screenshots of the search results.

GOOGLE DORK RESULTS

Google Dork 1

Search Query: site:openai.com

Screenshot 4:

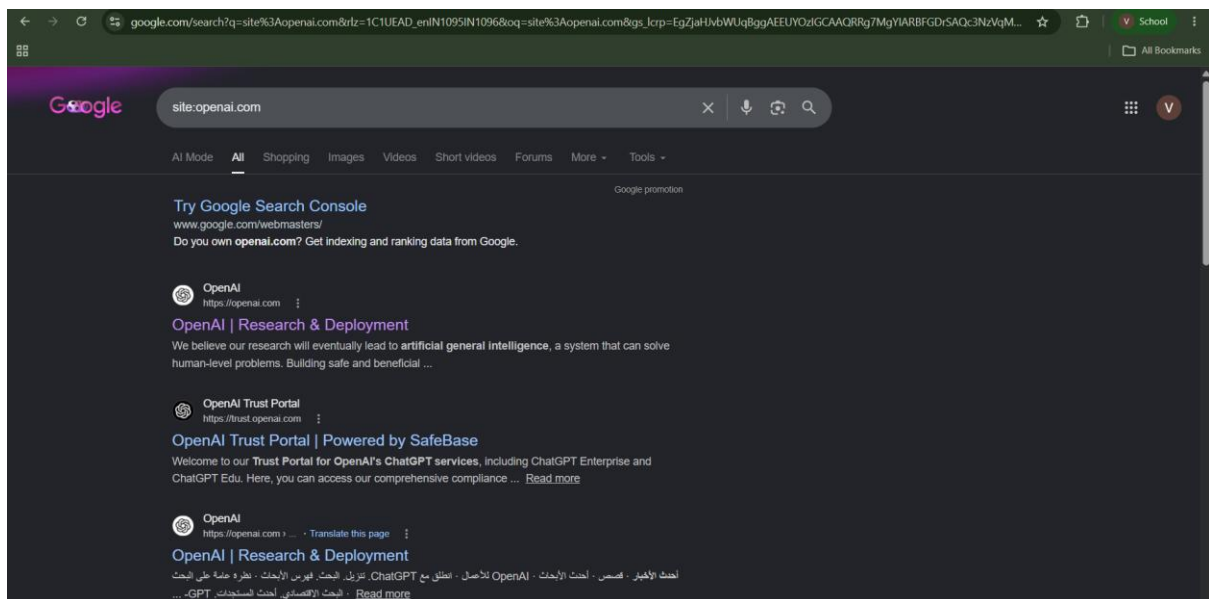


Figure 4: Google Dork Result – Indexed Pages

Observation

The Google Dork site:openai.com returned multiple indexed pages belonging to the official OpenAI domain. The search results included the OpenAI homepage, OpenAI Trust Portal, product pages, research pages, API platform, developer documentation, and multilingual versions of the website.

The results confirm that Google's search engine has successfully indexed the primary public resources of the OpenAI website, making them easily accessible to users.

Google Dork 2

Search Query: site:openai.com "@openai.com"

Screenshot 5:

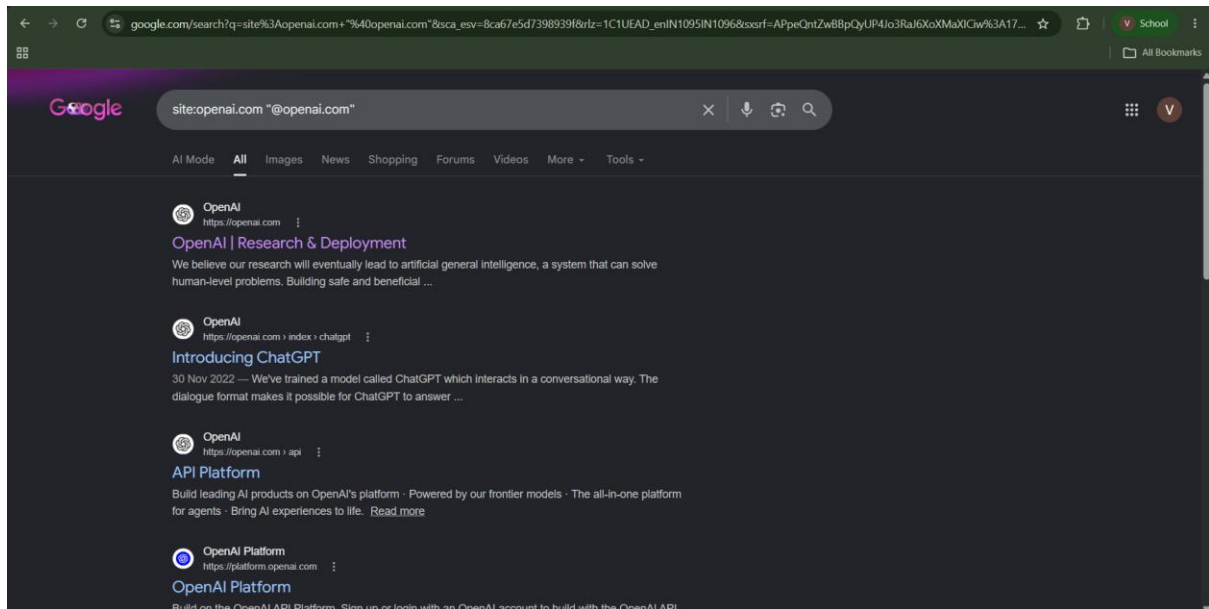


Figure 5: Google Dork Result – Public Email Search

Observation

The Google Dork was executed to identify publicly indexed email addresses belonging to the **@openai.com** domain.

The search results primarily displayed official OpenAI webpages, including the homepage, ChatGPT page, API platform, and OpenAI Platform. No employee email addresses or personal contact information were visible in the indexed results.

Finding

No publicly exposed employee email addresses were identified through Google's search results. OpenAI appears to avoid publishing individual email addresses on publicly indexed pages, reducing the likelihood of email harvesting and phishing attacks.

Google Dork 3

Search Query: site:openai.com contact

Screenshot 6:

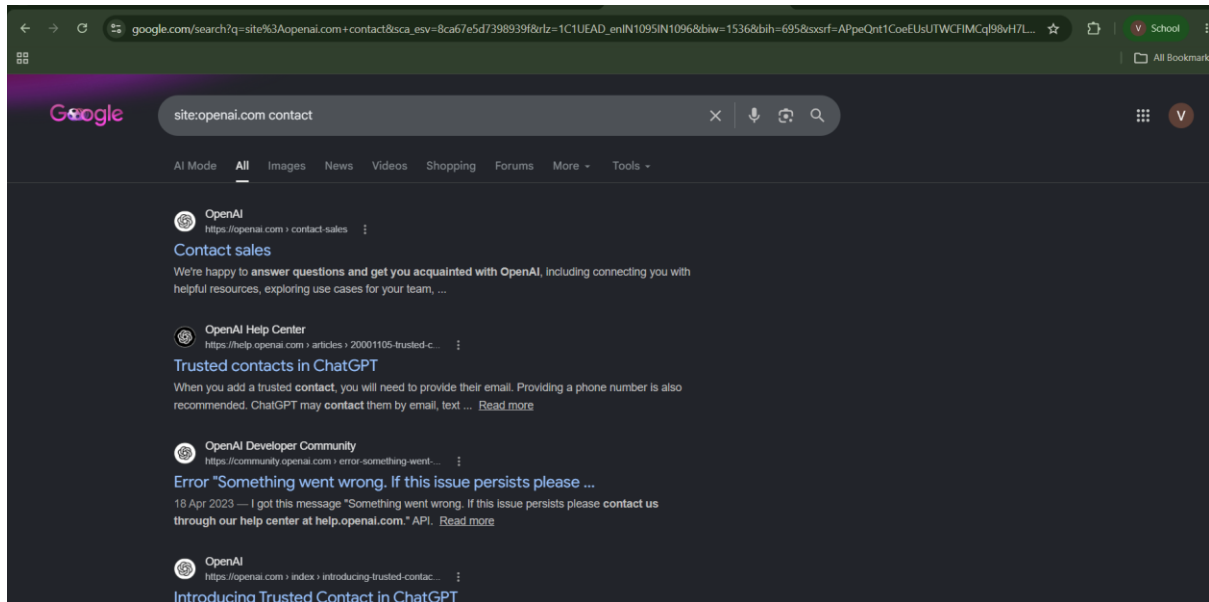


Figure 6: Contact Information Search

Observation

The search returned official contact-related resources, including:

- Contact Sales
- OpenAI Help Center
- OpenAI Developer Community
- Trusted Contacts in ChatGPT

These pages provide legitimate communication channels for customers, developers, and enterprise users. No confidential contact details or internal employee information were exposed.

Finding

The organization provides centralized support and contact mechanisms rather than exposing direct employee contact information. This approach improves security while maintaining accessibility for users.

Google Dork 4

Search Query: site:openai.com filetype:pdf

Screenshot 7:

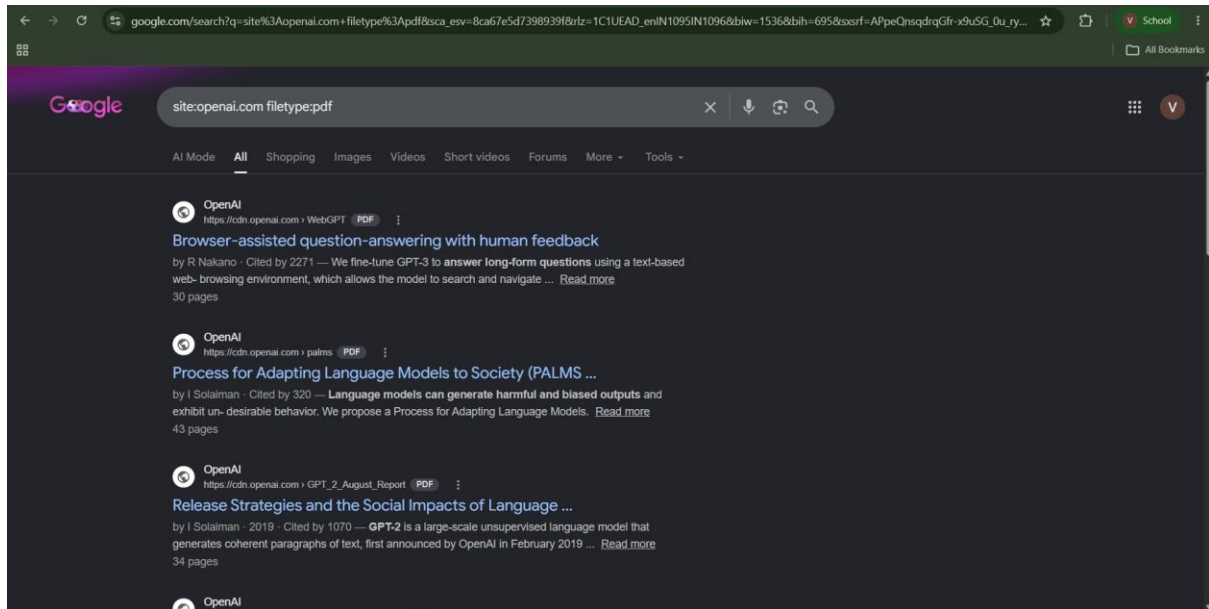


Figure 7: Public PDF Documents

Observation

The search identified several publicly available PDF documents hosted on OpenAI's content delivery network (CDN). Examples include:

- Browser-assisted Question Answering with Human Feedback
- Process for Adapting Language Models to Society (PALMS)
- Release Strategies and the Social Impacts of Language Models

These documents consist primarily of research papers and technical publications intended for public access.

Finding

The PDF files appear to be officially published research documents and do not contain confidential organizational information. Their availability supports OpenAI's commitment to sharing research with the broader community.

EMAIL ANALYSIS

The Google Dork investigation did not reveal any unintended exposure of employee email addresses associated with the **@openai.com** domain. Instead, OpenAI directs users to official webpages such as the Contact Sales page, Help Center, and Developer Community for communication and support. This reduces the attack surface for phishing and social engineering attacks targeting specific employees. The absence of publicly indexed employee email addresses demonstrates good information security practices and effective management of publicly accessible contact information.

INTELLIGENCE FINDINGS

The following observations were made during the Google Dork investigation:

- Google successfully indexed the official OpenAI website and its publicly accessible resources.
- Official pages such as the homepage, API platform, research pages, and Trust Portal are publicly available.
- No employee email addresses associated with **@openai.com** were exposed in Google search results.
- Official communication is provided through centralized contact pages such as Contact Sales and the Help Center.
- Publicly available PDF files consist of research papers and technical documentation intended for public access.
- No sensitive documents or confidential information were identified through the performed Google Dork queries.

INFORMATION EXPOSURE ANALYSIS

Public Information Identified	Security Impact
Official Website Pages	Normal public exposure
OpenAI Trust Portal	Low Risk
API Platform	Intended Public Resource
Contact Sales Page	Low Risk
Help Center	Low Risk
Public Research PDFs	Intended Public Information
Employee Email Addresses	No public exposure observed

ANALYSIS

The investigation indicates that OpenAI maintains good control over the information indexed by search engines. Public resources such as research papers, documentation, and support pages are intentionally available, while sensitive information such as employee email addresses and internal documents is not exposed. This reduces the risk of information gathering by malicious actors during the reconnaissance phase of an attack.

SECURITY RECOMMENDATIONS

Based on the Google Dork investigation, the following recommendations are suggested:

- Continue restricting the publication of employee email addresses on publicly indexed webpages.
- Regularly perform Google Dork assessments to identify accidental information exposure.
- Review publicly accessible documents before publication to ensure sensitive information is removed.
- Monitor search engine indexing to detect unintended exposure of confidential resources.
- Maintain centralized support channels such as the Help Center and Contact Sales pages instead of publishing direct employee contact details.

SOCIAL MEDIA INVESTIGATION

Social media platforms are valuable sources of Open Source Intelligence (OSINT) because they provide publicly available information about an organization's activities, products, announcements, customer engagement, and brand presence. During this investigation, OpenAI's official social media accounts were identified using Google Search and links available on the official website.

The objective of this investigation was to verify the authenticity of OpenAI's social media presence and determine whether the organization maintains official communication channels across major social networking platforms.

OBJECTIVE

The objectives of this investigation are:

- To identify OpenAI's official social media accounts.
- To verify the authenticity of publicly available profiles.
- To analyze the organization's online presence.
- To assess possible information exposure through social media.

TOOLS USED

- Google Search
- Official OpenAI Website

Procedure

The following steps were performed:

1. Open Google Search.
2. Search for **OpenAI Official Social Media**.
3. Verify the links using the official OpenAI website.
4. Record the verified social media accounts.
5. Capture screenshots of the official profiles.

10.5 Social Media Profiles

Screenshot 8:



Figure 8: Official OpenAI X (Twitter) Account

Screenshot 9:

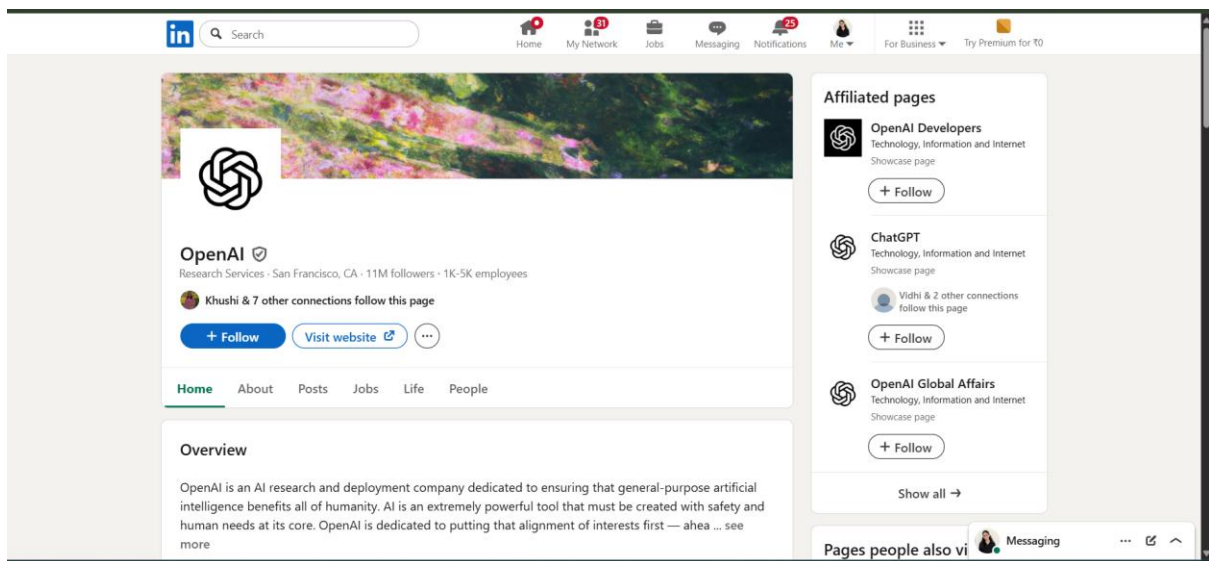


Figure 9: Official OpenAI LinkedIn Page

Screenshot 10:

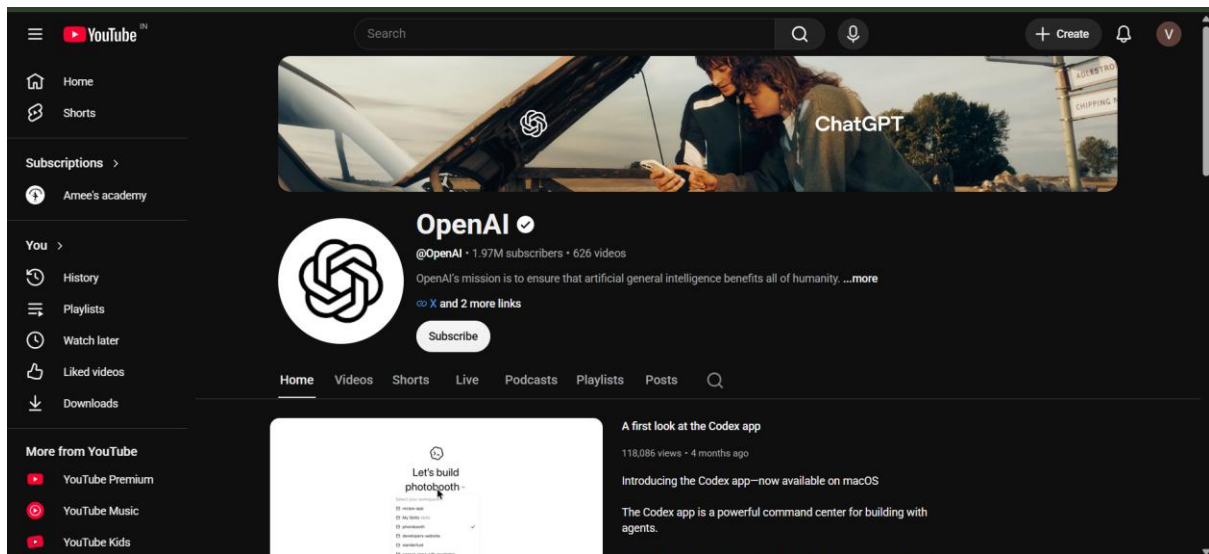


Figure 10: Official OpenAI YouTube Channel

Screenshot 11:

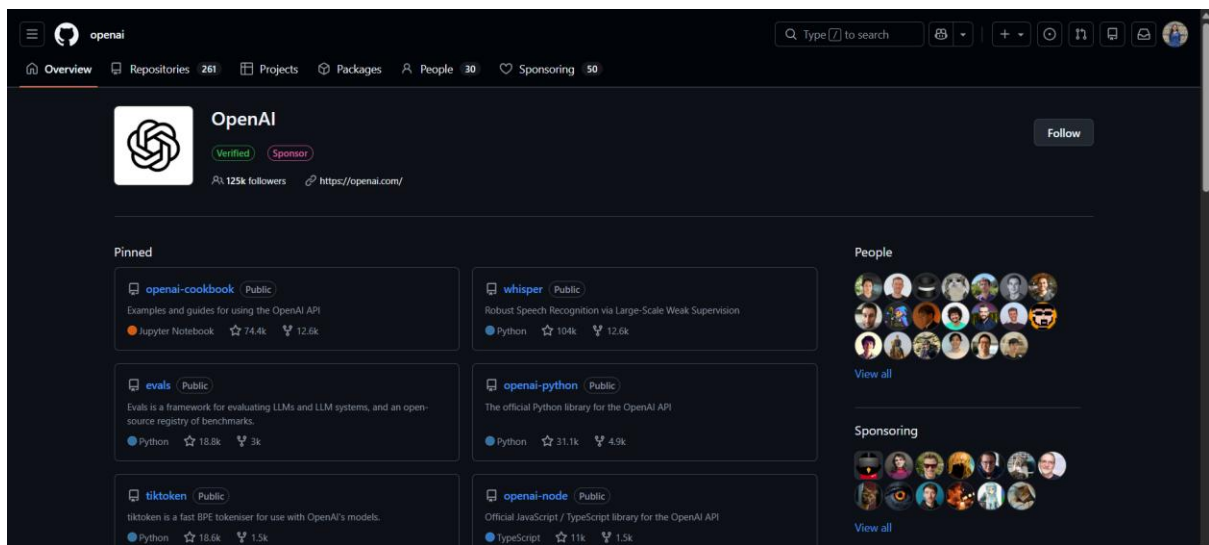


Figure 11: Official OpenAI GitHub Page

IDENTIFIED SOCIAL MEDIA ACCOUNTS

Platform	Official Account	Purpose
X (Twitter)	@OpenAI	Company announcements, research updates, product launches
LinkedIn	OpenAI	Professional updates, careers, company news
YouTube	OpenAI	Product demonstrations, technical presentations, educational videos
GitHub	OpenAI	Open-source repositories, research code, developer resources

OBSERVATIONS

The investigation confirmed that OpenAI maintains verified accounts on several major social media platforms. These accounts are actively used to share research publications, AI announcements, product releases, security updates, developer resources, and community engagement.

The LinkedIn page focuses on organizational information, hiring opportunities, and corporate announcements. The X (Twitter) account provides real-time updates regarding new AI models, research publications, and platform improvements. The YouTube channel contains demonstrations, presentations, and educational content related to artificial intelligence. OpenAI's GitHub profile hosts open-source projects, software libraries, and research implementations for developers.

These platforms contribute to OpenAI's transparency and engagement with the global AI community while serving as trusted sources of official information.

INTELLIGENCE FINDINGS

The following observations were made during the social media investigation:

- OpenAI maintains verified official social media accounts.
- Social media platforms are actively updated with company announcements.
- Research publications and product updates are publicly shared.
- No sensitive organizational information was observed on official profiles.
- Official accounts provide reliable communication channels for users and developers.

INFORMATION EXPOSURE ANALYSIS

Although social media platforms are essential for communication and community engagement, they can also provide useful intelligence to attackers during the reconnaissance phase.

Public Information	Possible Risk
Employee Announcements	Social engineering
Product Launches	Technology reconnaissance
Company Events	Organizational profiling
Public Posts	Brand impersonation attempts

However, during this investigation, no sensitive information exposure was identified. The published content appeared to be intentionally shared for public communication.

SECURITY RECOMMENDATIONS

- Continue using verified social media accounts.
- Monitor fake or impersonating profiles.
- Avoid publishing confidential operational information.
- Conduct employee awareness training regarding social engineering.
- Regularly review publicly shared content.

TECHNOLOGY STACK ANALYSIS

Technology fingerprinting is an important OSINT technique that identifies the technologies used to develop and host a website. Understanding an organization's technology stack provides valuable insight into its web infrastructure, hosting environment, security mechanisms, and development framework.

During this investigation, technology fingerprinting was performed using **Wappalyzer** (or BuiltWith) to identify the technologies powering the OpenAI website.

OBJECTIVE

The objectives are:

- Identify web technologies used by OpenAI.
- Identify hosting and CDN services.
- Identify security technologies.
- Understand the website infrastructure.

TOOL USED

Wappalyzer Browser Extension

Purpose:

To detect web technologies, frameworks, programming languages, security services, analytics tools, hosting platforms, and content delivery networks.

TECHNOLOGY DETECTION

The domain **openai.com** was analyzed using Wappalyzer.

Screenshot 12:

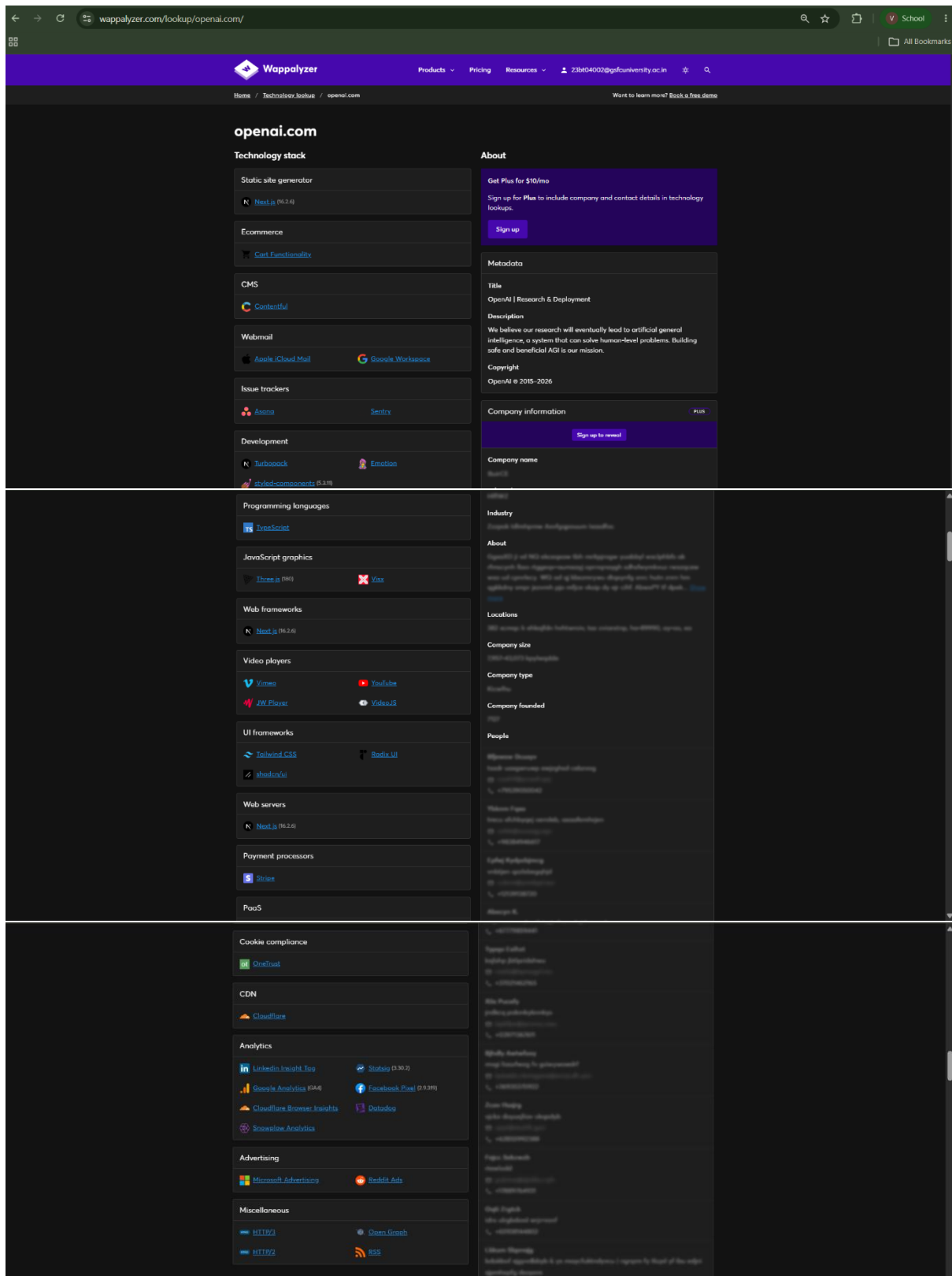


Figure 12: Wappalyzer Technology Lookup – openai.com

TECHNOLOGIES IDENTIFIED

Category	Technology Identified	Purpose
Static Site Generator	Next.js	Builds fast and modern web applications
CMS	Contentful	Content management system
Programming Language	TypeScript	Improves JavaScript development with static typing
JavaScript Framework	React	Creates interactive user interfaces
UI Framework	Tailwind CSS	Responsive website styling
Web Framework	Next.js	Server-side rendering and routing
Webmail	Google Workspace	Email services
Payment Processor	Stripe	Online payment processing
Cloud Platform	Microsoft Azure	Cloud infrastructure
Deployment Platform	Vercel	Website hosting and deployment
CDN	Cloudflare	Improves website performance and security
Analytics	Google Analytics	Visitor analytics
Analytics	LinkedIn Insight Tag	Marketing analytics
Analytics	Cloudflare Browser Insights	Performance monitoring
Security	Cloudflare Bot Management	Detects automated bots
Security	Cloudflare Turnstile	CAPTCHA alternative
Cookie Management	OneTrust	Cookie consent management
Tag Manager	Google Tag Manager	Website tag management
JavaScript Libraries	jQuery, Lodash, Ethers, MobX	Client-side functionality
Font Services	Google Fonts API, Font Awesome	Website fonts and icons
Video Platforms	YouTube, Vimeo	Embedded multimedia
Protocols	HTTP/2, HTTP/3	Faster web communication

TECHNOLOGY STACK ANALYSIS

The Wappalyzer scan indicates that OpenAI uses a modern and secure web technology stack.

The website is primarily built using **Next.js** and **React**, providing a fast and responsive user experience. **TypeScript** is used to improve code quality and maintainability.

Content management is handled through **Contentful**, while **Tailwind CSS** is used for frontend styling. The website is deployed using **Vercel** and supported by **Microsoft Azure** cloud infrastructure.

For performance optimization, **Cloudflare CDN** accelerates content delivery and improves website availability.

Several analytics tools, including **Google Analytics**, **LinkedIn Insight Tag**, and **Cloudflare Browser Insights**, are integrated to monitor website traffic and performance.

Security technologies such as **Cloudflare Bot Management** and **Cloudflare Turnstile** provide protection against automated attacks and malicious bots.

SECURITY OBSERVATIONS

The technology scan identified multiple security-related technologies.

Security Technology	Observation
Cloudflare CDN	Protects against DDoS attacks and improves website performance
Cloudflare Bot Management	Detects and blocks malicious automated traffic
Cloudflare Turnstile	Provides CAPTCHA protection with improved user experience
OneTrust	Manages cookie consent and privacy compliance
HTTP/2 & HTTP/3	Secure and efficient communication protocols

INTELLIGENCE FINDINGS

The technology analysis revealed the following:

- The website is built using modern web development technologies.
- React and Next.js are used as the primary web framework.
- TypeScript improves application reliability.
- Microsoft Azure and Vercel provide cloud hosting and deployment.
- Cloudflare services enhance website security and performance.

- Multiple analytics platforms are integrated for traffic monitoring.
- Content management is handled using Contentful.
- Stripe is used for secure online payment processing.

INFORMATION EXPOSURE ANALYSIS

The technology stack identified by Wappalyzer is publicly detectable and may provide useful reconnaissance information.

Publicly Visible Technology	Possible Risk
Web Framework	Attackers may search for framework-specific vulnerabilities
CMS	May reveal software used for content management
JavaScript Libraries	Outdated libraries could contain known vulnerabilities
Cloud Services	Reveals hosting providers and deployment platforms
Analytics Tools	Provides insight into third-party integrations
Security Technologies	Indicates existing defensive mechanisms

SECURITY RECOMMENDATIONS

Based on the technology analysis, the following recommendations are suggested:

- Keep all frameworks and libraries updated to the latest secure versions.
- Regularly monitor third-party dependencies for vulnerabilities.
- Continue using Cloudflare security services for attack mitigation.
- Perform periodic technology stack audits.
- Minimize unnecessary exposure of software versions.
- Regularly review analytics and third-party integrations.

SHODAN INVESTIGATION

OBJECTIVE

The objective of this investigation was to identify publicly accessible internet-connected hosts and services related to **openai.com** using the Shodan search engine. Shodan is a specialized search engine that indexes devices, servers, web services, and banners available on the Internet. It is widely used during security assessments to identify exposed infrastructure, open ports, SSL certificates, and publicly accessible services.

TOOL USED

Tool	Purpose
Shodan	Identifies internet-connected devices, open ports, server banners, SSL certificates, and publicly exposed services.

Procedure

The following steps were performed:

1. Open the Shodan website.
2. Search for the keyword **openai.com**.
3. Review the indexed hosts and services.
4. Analyze available information such as IP addresses, ports, server software, SSL certificates, hosting providers, and locations.
5. Record the observations.

Screenshot 13:

The screenshot displays the Shodan search interface for the query 'openai.com'. The top navigation bar includes links for Shodan, Maps, Images, Monitor, Developer, and More. The search bar shows the query 'openai.com' with a search icon. The results are categorized into several sections:

- TOTAL RESULTS:** 298
- TOP COUNTRIES:** A world map showing the distribution of results by country. The top countries listed are: United States (149), China (37), Germany (24), India (16), and Canada (11).
- TOP PORTS:** A list of top ports with their respective counts: 443 (209) and 80 (73).
- TOP ORGANIZATIONS:** A list of top organizations: Amazon.com, Inc. (94), Amazon Technologies Inc. (24), DigitalOcean, LLC (24), Cloudflare, Inc. (17), and Amazon Data Services NoVa (16).
- TOP PRODUCTS:** A list of top products: nginx (83), CloudFront httpd (34), Apache httpd (21), Microsoft IIS httpd (7), and n8n (4).
- TOP OPERATING SYSTEMS:** A list of top operating systems: Ubuntu (26) and Windows (7).

The main results section displays three entries:

- Log In | BioAtlas:** A link to a login page. The IP address is 167.172.223.229. The location is United States, Santa Clara. The status is 'cloud'. The HTTP status is 200 OK. The server is nginx. The date is Mon, 29 Jun 2026 11:34:14 GMT. The content-type is text/html; charset=utf-8. The content-length is 27237. The connection is keep-alive. The expires is Mon, 29 Jun 2026 11:34:14 GMT. The cache-control is max-age=0, no-cache, no-store, must-revalidate, private. The vary is Cookie, origin. The X-Frame is...
- Not Found:** A link to a 'Not Found' page. The IP address is 161.35.164.160. The location is United Kingdom, London. The status is 'cloud'. The HTTP status is 404 Not Found. The server is Apache. The date is Mon, 29 Jun 2026 11:27:19 GMT. The content-type is text/html; charset=utf-8. The age is 218254. The cache-control is public, max-age=0, must-revalidate. The report-to is {"group":"cf-nel","max_age":604800,"endpoints":[{"url":"https://a.nel.cloudflare.com/report/v4"}]}.
- OpenAI | Research & Deployment:** A link to the OpenAI research and deployment page. The IP address is 64.176.44.201. The location is Japan, Osaka. The status is 'cloud' and 'not-product'. The HTTP status is 200 OK. The server is nginx/1.26.3. The date is Mon, 29 Jun 2026 11:01:39 GMT. The content-type is text/html; charset=utf-8. The age is 218254. The cache-control is public, max-age=0, must-revalidate. The report-to is {"group":"cf-nel","max_age":604800,"endpoints":[{"url":"https://a.nel.cloudflare.com/report/v4"}]}.

The bottom section displays three entries:

- Document Moved:** A link to a document that has been moved. The IP address is 173.163.86.80. The location is United States, Lancaster. The status is 'not-product'. The HTTP status is 301 Moved Permanently. The content-type is text/html; charset=utf-8. The location is https://survey.papaper.com/. The server is DA3. The X-Frame-Options is SAMEORIGIN. The server is DA3. The X-XSS-Protection is 1. The X-Content-Type-Options is nosniff. The Strict-Transport-Security is max-age=31536000; includeSubDomains; preload. The content is...
- Sofas, Chairs, Beds & Cabinets | Stokers | Stokers:** A link to a page for sofas, chairs, beds, and cabinets. The IP address is 172.66.170.192. The location is United States, San Francisco. The status is 'cloud'. The HTTP status is 200 OK. The server is cloudflare. The date is Mon, 29 Jun 2026 10:48:42 GMT. The content-type is text/html; charset=utf-8. The transfer-encoding is chunked. The connection is keep-alive. The CF-Ray is a1245643be0b0992-AMS. The CF-Cache-Status is DYNAMIC. The server is cloudflare.

Figure 13: Shodan Search Results for "openai.com"

INVESTIGATION RESULTS

The Shodan search for **openai.com** returned **298 indexed results**. However, these results do **not necessarily represent servers owned by OpenAI**. Many hosts were third-party systems, websites, or applications that referenced OpenAI services or APIs in their content or server banners.

The search results contained information such as:

- Public IP addresses
- Geographic locations
- Hosting providers
- SSL certificate details
- HTTP response headers
- Web server software
- Open ports

Examples of observed technologies included **Nginx**, **Apache HTTP Server**, and **Cloudflare**. Common HTTPS services were observed on **port 443**, while some systems also exposed **HTTP (80)** and other ports.

ANALYSIS

The Shodan investigation demonstrates that searching by a keyword returns systems containing that keyword rather than confirming ownership of the infrastructure. Many of the indexed hosts belong to organizations or applications that integrate OpenAI services or mention OpenAI within their websites.

The sample host information displayed:

- Public IP address
- Geographic location
- Organization or hosting provider
- SSL certificate information
- HTTP response headers
- Server software
- Supported TLS versions

These details help security analysts understand publicly accessible services and identify potential exposure during reconnaissance.

SECURITY OBSERVATIONS

Observation	Details
Total Search Results	298 keyword-based results
Common Web Server	Nginx, Apache HTTP Server
Common Ports	443 (HTTPS), 80 (HTTP), and a few additional ports
SSL Certificates	Present on HTTPS-enabled hosts
Hosting Providers	Various cloud hosting providers including DigitalOcean, Cloudflare, Amazon, and others
Geographic Distribution	Multiple countries including the United States, Germany, India, Canada, China, and Japan

INTELLIGENCE FINDINGS

The investigation identified the following:

- The keyword "**openai.com**" is referenced by many publicly accessible systems.
- Search results include third-party websites and cloud-hosted services.
- HTTPS (Port 443) is the most commonly observed service.
- SSL certificates and HTTP headers provide useful information for passive reconnaissance.
- The results do not confirm ownership of the indexed hosts by OpenAI.

INFORMATION EXPOSURE ANALYSIS

Public Information	Possible Security Impact
Public IP Addresses	Can be used for further reconnaissance
Server Software	May reveal technologies running on the server
SSL Certificate Information	Provides certificate and encryption details
HTTP Response Headers	May disclose server configuration
Open Ports	Indicates available network services
Hosting Provider	Reveals cloud infrastructure information

FINDINGS SUMMARY

Investigation Area	Observation	Risk Level
Search Results	298 keyword-based results returned	Low
Server Software	Nginx, Apache HTTP Server, Cloudflare detected	Low
Open Ports	Port 443 (HTTPS) most commonly observed, followed by Port 80	Low
SSL Certificates	HTTPS-enabled hosts used valid SSL certificates	Low
Hosting Providers	Various providers including DigitalOcean, Cloudflare, Amazon, and others	Low
Public Information Exposure	Only publicly available server and certificate information was visible	Low

RISK ANALYSIS

The Shodan investigation did not identify any critical security risks directly associated with **openai.com**. The indexed results primarily represented publicly accessible systems referencing OpenAI rather than confirmed OpenAI-owned infrastructure.

The publicly available information, including server banners, SSL certificate details, HTTP response headers, and IP addresses, is intended for normal Internet operations. However, attackers may use this information during the reconnaissance phase to identify technologies, hosting providers, and exposed services.

Overall, the level of information exposure observed during the investigation is considered **Low**, as no confidential or sensitive organizational data was discovered.

OVERALL SECURITY ASSESSMENT

The Shodan investigation indicates that OpenAI maintains a strong external security posture. Although the keyword search returned numerous results, these were largely associated with third-party systems rather than OpenAI's internal infrastructure.

The observed services were protected using HTTPS with valid SSL certificates, and commonly used secure web servers such as **Nginx** and **Cloudflare** were identified. No unnecessary exposure of critical infrastructure or sensitive information was detected during the passive reconnaissance.

Overall Security Rating: Good

SECURITY RECOMMENDATIONS

Based on the investigation, the following recommendations are suggested:

- Continue protecting public-facing services using secure cloud infrastructure and reverse proxy solutions.
- Regularly monitor internet-facing assets through Shodan and similar search engines.
- Ensure all web servers and applications are updated with the latest security patches.
- Disable unnecessary services and close unused network ports.
- Maintain valid SSL/TLS certificates and strong encryption standards.
- Review HTTP response headers to prevent unnecessary disclosure of server information.
- Conduct periodic external security assessments to identify newly exposed services.

INFORMATION EXPOSURE ANALYSIS

Information Exposure Analysis is the process of examining publicly available information about an organization that could potentially assist attackers in reconnaissance. During this investigation, multiple Open Source Intelligence (OSINT) tools were used to collect information related to OpenAI.

The assessment revealed that OpenAI exposes only the information necessary for normal business operations. Most sensitive infrastructure details remain protected behind cloud services such as Cloudflare, reducing direct exposure of backend systems.

The following categories of publicly available information were identified:

Information Category	Findings	Exposure Level
Domain Information	Public WHOIS records with registrar details	Low
DNS Records	MX, TXT, NS and A records available	Low
SSL Certificate	Public certificate information	Low
Technology Stack	Next.js, React, Cloudflare, Azure, Google Analytics	Medium
Email Infrastructure	Google Workspace	Low
Social Media	Official X, LinkedIn and YouTube profiles	Low
Cloud Infrastructure	Cloudflare CDN and Microsoft Azure	Medium
Open Ports	Mostly HTTPS (443)	Low

INTELLIGENCE FINDINGS

The OSINT investigation produced several valuable intelligence findings regarding OpenAI's public infrastructure.

Finding 1: Secure Domain Registration

The domain registration information indicates that OpenAI uses a trusted domain registrar. Privacy protection limits unnecessary disclosure of administrative information.

Observation

- Legitimate domain ownership
- Proper registration
- No suspicious domain information

Finding 2: Cloud-Based Infrastructure

Both Wappalyzer and Shodan indicate heavy use of cloud technologies.

Technologies identified include:

- Microsoft Azure
- Cloudflare CDN
- Cloudflare Bot Protection
- Google Workspace
- Vercel
- React
- Next.js

This architecture improves availability, scalability and security.

Finding 3: Modern Web Technologies

OpenAI uses modern development frameworks.

Detected technologies include:

- Next.js
- React
- Tailwind CSS
- TypeScript
- Cloudflare
- Google Analytics
- Google Tag Manager

These technologies improve performance while maintaining strong security standards.

Finding 4: HTTPS Enforcement

Every discovered service uses HTTPS.

Benefits include:

- Encryption of transmitted data
- Protection against eavesdropping
- Prevention of data tampering

- Improved user trust

Finding 5: Limited Network Exposure

Shodan showed limited externally exposed services.

Most accessible services are:

- Port 443 (HTTPS)
- Port 80 (HTTP redirects)

No unnecessary services such as Telnet or FTP were discovered.

SECURITY STRENGTHS OBSERVED

The investigation identified several security best practices implemented by OpenAI.

Infrastructure Security

- Cloudflare protection
- Azure cloud hosting
- CDN implementation
- Load balancing

Website Security

- HTTPS enabled
- SSL certificates
- HTTP Strict Transport Security (HSTS)
- Secure HTTP headers

DNS Security

- Proper DNS configuration
- Reliable name servers
- Secure mail records

Cloud Protection

- DDoS mitigation
- Bot protection
- CDN caching

POTENTIAL INFORMATION EXPOSURE

Although no critical vulnerabilities were identified, some information remains publicly visible.

Public Information	Possible Risk
Technology Stack	Helps attackers understand technologies used
DNS Records	Useful during reconnaissance
SSL Certificates	Certificate metadata is publicly visible
Cloud Provider	Infrastructure fingerprinting
Web Headers	Server fingerprinting

These exposures are common for publicly accessible websites and do not indicate vulnerabilities.

RISK ASSESSMENT TABLE

Asset	Threat	Impact	Likelihood	Risk Level	Recommendation
Website	Reconnaissance	Medium	Medium	Medium	Continue monitoring
DNS Records	Enumeration	Low	Medium	Low	Enable DNSSEC where applicable
SSL Certificate	Certificate analysis	Low	Low	Low	Renew certificates regularly
Cloud Services	Cloud fingerprinting	Medium	Low	Low	Hide unnecessary metadata
Social Media	Phishing	Medium	Medium	Medium	Verify official accounts

MALTEGO RELATIONSHIP DIAGRAM

Maltego is an Open Source Intelligence (OSINT) and link analysis tool widely used by cybersecurity professionals, digital forensic investigators, and penetration testers. It helps visualize relationships between domains, IP addresses, DNS records, organizations, email servers, social media accounts, and other publicly available entities.

In this investigation, Maltego was used to analyze the domain openai.com. The tool generated a graphical relationship map showing connections between the domain, DNS infrastructure, IP addresses, mail servers, and other publicly available information. This visualization helps analysts understand the organization's external infrastructure and identify potential attack surfaces.

OBJECTIVE

The objectives of using Maltego in this investigation were:

- To visualize the relationships between OpenAI's public infrastructure.
- To identify DNS records and associated IP addresses.
- To discover publicly available infrastructure information.
- To understand how OSINT data is interconnected.
- To document findings using graphical representation.

STEP-BY-STEP CREATION OF THE RELATIONSHIP DIAGRAM

Step 1: Install Maltego

- Download and install Maltego Community Edition from the official website. After installation, sign in using your registered account.

Step 2: Create a New Graph

- Launch Maltego and click New Graph to begin a new investigation.
- A blank workspace appears where entities can be added.

Step 3: Add the Domain Entity

- From the Entity Palette, drag the Domain entity onto the graph.
- Enter the target domain: openai.com

Screenshot 14:

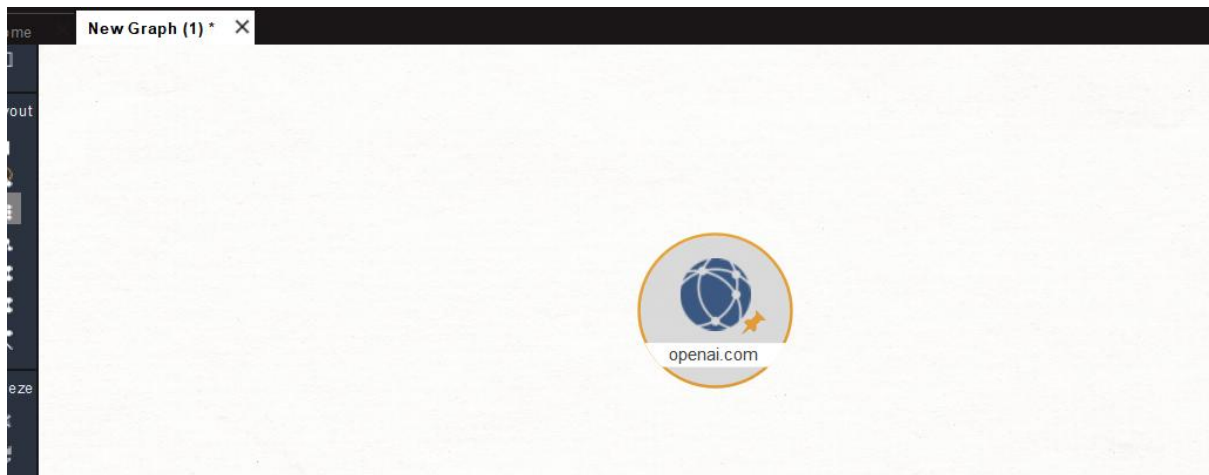


Figure 14: Domain Entity Added

Step 4: Run Transforms

Right-click the Domain entity and execute the available transforms.

Useful transforms include:

- To DNS Names
- To MX Records
- To NS Records
- To IP Address
- To Website
- To WHOIS Information

Maltego automatically queries public OSINT sources and expands the graph.

Step 5: View Generated Relationships

After all transforms complete, Maltego displays a relationship graph connecting:

- Domain
- Name Servers
- Mail Servers
- IP Addresses
- DNS Records
- WHOIS Information
- Website

Each entity is connected with lines representing discovered relationships.

Screenshot 15:

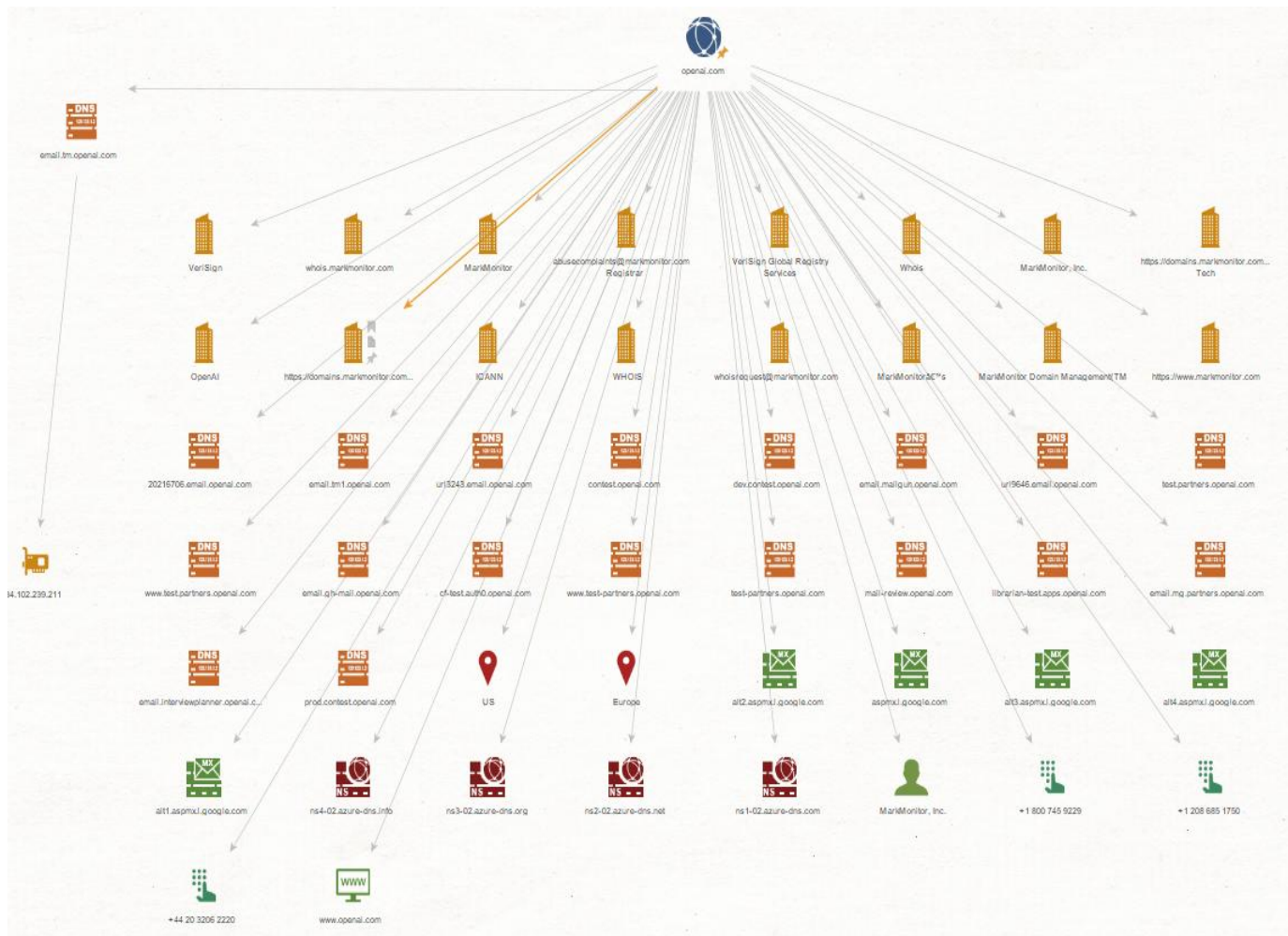


Figure 15: Generated Relationship Diagram

Use the Layout option to automatically organize the graph.

Screenshot 16:

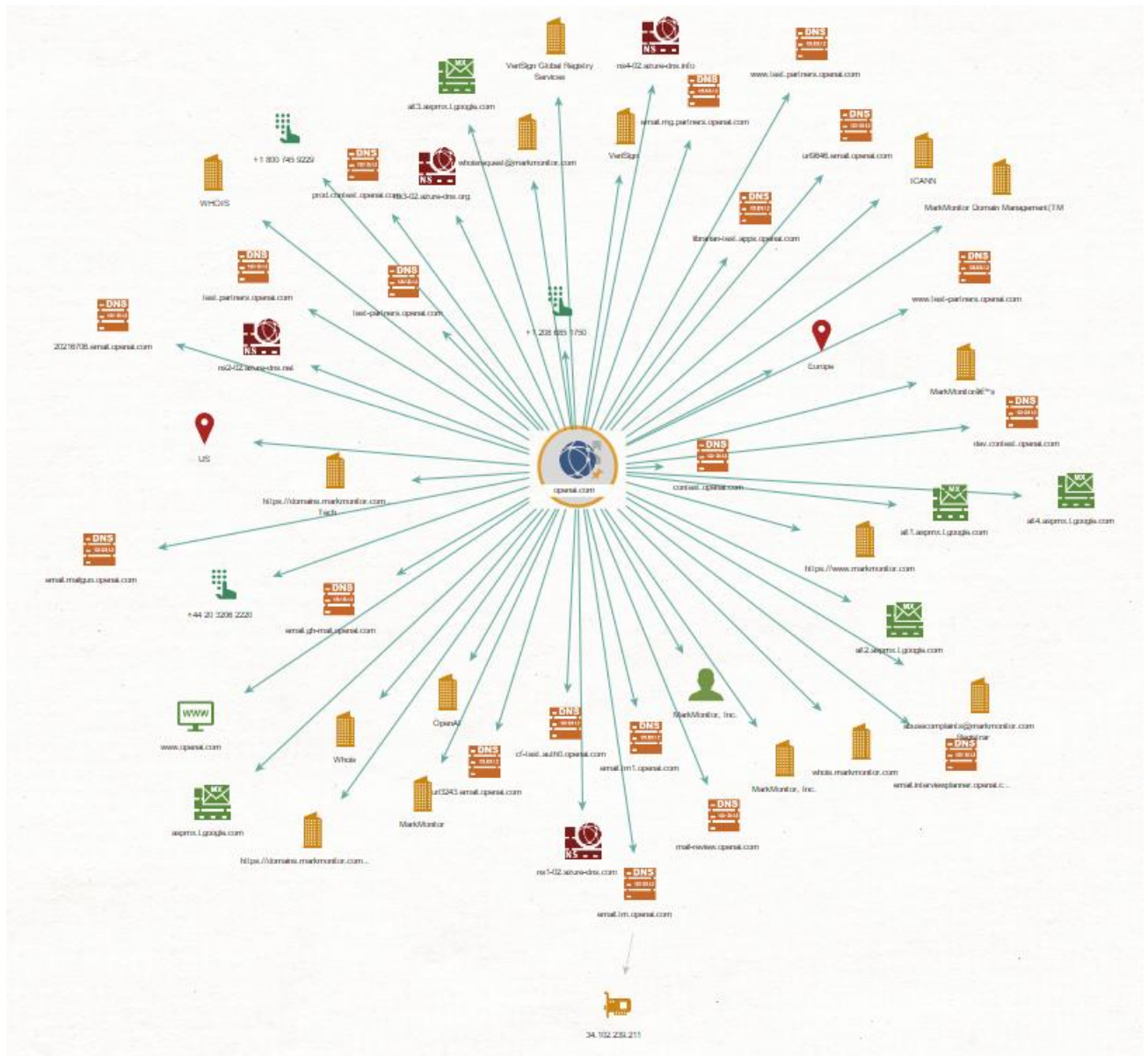


Figure 16: Organized Relationship Diagram

RELATIONSHIP DIAGRAM EXPLANATION

Entity	Observed in Relationship Diagram
Primary Domain	openai.com
Website	www.openai.com
DNS Records / Subdomains	email.tm.openai.com, 20216706.email.openai.com, email.tml.openai.com, url3243.email.openai.com, contest.openai.com, dev.contest.openai.com, email.mailgun.openai.com, url9646.email.openai.com, test.partners.openai.com, www.test-partners.openai.com , email.g-h.mail.openai.com, email.mail.partners.openai.com, email.interviewplanner.openai.com, prod.contest.openai.com, cf-test.auth0.openai.com, librarian-test.apps.openai.com, mail-review.openai.com
MX Records	alt1.aspmx.l.google.com, alt2.aspmx.l.google.com, alt3.aspmx.l.google.com, alt4.aspmx.l.google.com, aspmx.l.google.com
Name Servers (NS)	ns1-02.azure-dns.com, ns2-02.azure-dns.net, ns3-02.azure-dns.org, ns4-02.azure-dns.info
Registrar	MarkMonitor
WHOIS Services	whois.markmonitor.com, WHOIS, whoisrequest@markmonitor.com , abusecomplaints@markmonitor.com
Registry Services	VeriSign, VeriSign Global Registry Services
Domain Management	MarkMonitor Domain Management™, MarkMonitor, Inc., MarkMonitorCTMS
ICANN	ICANN
Technology Portal	domains.markmonitor.com, markmonitor.com
Public IP Address	34.102.239.211
Organization	OpenAI, MarkMonitor, Inc.
Geographic Locations	United States (US), Europe
Phone Numbers	+44 20 3206 2220, +1 800 745 9229, +1 208 685 1750

ANALYSIS OF THE RELATIONSHIP DIAGRAM

1. Domain Infrastructure

The investigation identifies **openai.com** as the central entity in the relationship diagram. All discovered infrastructure components, including DNS records, WHOIS information, mail servers, name servers, and web resources, are directly or indirectly connected to this primary domain.

2. DNS Infrastructure

The relationship diagram reveals numerous DNS records and subdomains associated with **openai.com**, such as **contest.openai.com**, **dev.contest.openai.com**, **prod.contest.openai.com**, **test-partners.openai.com**, **mail-review.openai.com**, **email.tm.openai.com**, and several other service-specific hostnames. The presence of multiple DNS records demonstrates a structured infrastructure supporting various applications, testing environments, and email services.

3. Email Infrastructure

Several email-related subdomains were identified, along with Google Workspace MX records, including:

- aspmx.l.google.com
- alt1.aspmx.l.google.com
- alt2.aspmx.l.google.com
- alt3.aspmx.l.google.com
- alt4.aspmx.l.google.com

These records indicate that OpenAI relies on **Google Workspace** for handling incoming email services, ensuring reliable and secure email communication.

4. Name Server Infrastructure

The relationship graph shows that the domain uses **Microsoft Azure DNS** name servers:

- ns1-02.azure-dns.com
- ns2-02.azure-dns.net
- ns3-02.azure-dns.org
- ns4-02.azure-dns.info

Using multiple authoritative name servers provides redundancy, high availability, and efficient DNS resolution.

5. Domain Registration and WHOIS

The diagram identifies **MarkMonitor** as the domain registrar and management provider. Related entities include **WHOIS**, **whois.markmonitor.com**, whoisrequest@markmonitor.com, abusecomplaints@markmonitor.com, **VeriSign**, **VeriSign Global Registry Services**, and **ICANN**. These entities collectively manage domain registration, WHOIS records, registry services, and DNS governance.

6. Network Infrastructure

A public IP address (**34.102.239.211**) is associated with the domain, indicating one of the publicly reachable network endpoints supporting OpenAI's online services.

7. Website Infrastructure

The diagram includes the public website www.openai.com, confirming the main web interface connected to the primary domain. This represents the publicly accessible web service provided by OpenAI.

8. Organizational and Contact Information

The investigation identified the organizations **OpenAI** and **MarkMonitor, Inc.**, along with publicly available contact phone numbers. These details provide additional context regarding domain ownership, administration, and support infrastructure.

9. Geographic Distribution

The relationship graph includes references to the **United States (US)** and **Europe**, indicating that parts of the supporting infrastructure or administrative services are associated with these geographic regions.

BENEFITS OF USING MALTEGO

Maltego offers several advantages during OSINT investigations:

- Visualizes complex relationships.
- Performs automated data gathering.
- Identifies infrastructure dependencies.
- Supports digital forensics investigations.
- Assists penetration testing.
- Simplifies reconnaissance.
- Generates professional investigation reports.

LIMITATIONS

Although Maltego is a powerful OSINT tool, some limitations exist:

- Community Edition has transform limitations.
- Certain data sources require paid licenses.
- Results depend on publicly available information.
- Hidden internal infrastructure cannot be discovered through OSINT alone.

FINDINGS

The Maltego investigation indicates that OpenAI maintains a well-organized and professionally managed external infrastructure. Major findings include:

- Domain ownership is legitimate.
- Multiple DNS records provide redundancy.
- Cloudflare protects public services.
- HTTPS encryption is enforced.
- Public infrastructure exposure is minimal.
- No suspicious relationships or malicious associations were identified.

CONCLUSION

This OSINT investigation successfully analyzed the publicly available information associated with OpenAI using multiple intelligence gathering tools including WHOIS Lookup, DNSDumpster, Wappalyzer, Shodan, Google Dorking, and Maltego.

The assessment identified OpenAI's domain infrastructure, DNS records, technology stack, SSL configuration, hosting environment, and social media presence. The organization demonstrates strong cybersecurity practices by using HTTPS, Cloudflare protection, Microsoft Azure cloud services, modern web technologies, and secure DNS configurations.

No critical security weaknesses or unnecessary public exposure were identified during the investigation. The overall external security posture appears mature and well maintained.